Compiling FFI-heavy OCaml to WebAssembly

An Experience Report on MOPSA

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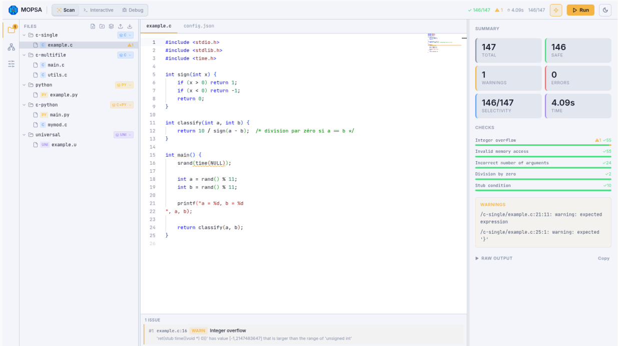
The project

MOPSA: a static analyzer by abstract interpretation, for C & Python

- written *mostly* in OCaml...
- ...on GMP, MPFR, Zarith, Apron, LLVM/Clang. All bound through the FFI



Goal: run it entirely in the browser. No server.

It works today: https://mopsawasm.rboud.com/

Ship the runtime itself

The missing `caml_*` symboles are defined in the **the OCaml runtime**.

1. compile MOPSA to **bytecode**
2. compile the **runtime** + every native dep to wasm (Emscripten)
3. **link statically** into one module
4. **interpret** `mopsa.bc` on top

```
┌──────────────────────────────┐
│   mopsa.bc  (interpreted)    │
├──────────────────────────────┤
│ ocamlrun.wasm · one module   │
│ libcamlrun + prims           │
│ GMP · MPFR · Apron · Zarith  │
│ camlidl rt · LLVM/Clang      │
│ Clang_to_ml                  │
└──────────────────────────────┘
```

One toolchain (emcc), one memory model, no per-binding glue.

Why not x ?

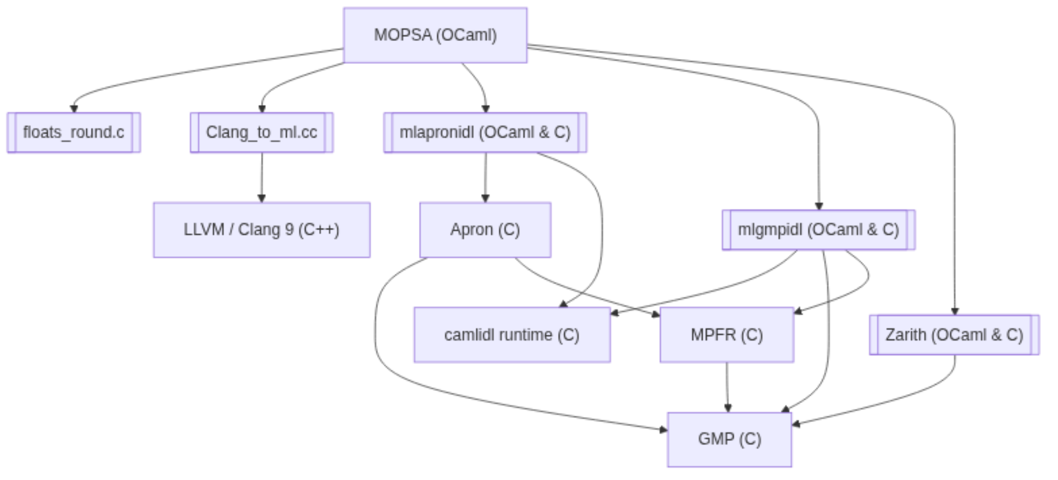
`js_of_ocaml`, `wasm_of_ocaml`, `wasocaml`: they compile the OCaml and **leave the C/C++ behind**.

- fine for pure OCaml, or with a JS reimplementation (`zarith_stubs_js`)
- here: rewrite 5000 lines of `Clang_to_ml.cc` *and* 655 generated stubs?

A structural problem: **what do the FFI stubs link against?** `caml_alloc`, `caml_callback`, … exist in no translated module.

And exposing them wouldn't help: WasmGC backends move OCaml values *out of linear memory*, so a C stub reading a tag by pointer cannot reach them at all.

The dependency stack



Every box ships C or C++ · double-bordered boxes use the FFI to build or read OCaml values

The FFI goes both ways

`Clang_to_ml.cc` 5000 lines, both worlds in one translation unit:

```
#include "clang/AST/RecursiveASTVisitor.h"
#include "clang/Frontend/CompilerInstance.h"

#include <caml/mlvalues.h>
#include <caml/alloc.h>
#include <caml/memory.h>
```

Each Clang AST node triggers `caml_alloc` + `Store_field`: allocation *inside the OCaml GC heap, from C++*.

CamlIDL generates ~655 more such stubs for Apron & GMP.

Every call site bakes in what an OCaml value is, byte for byte, in memory.

Binding 1435 externals without dlopen

Bytecode never calls C directly: `C_CALLn <index>`, resolved at startup by `dlopen`/`dlsym`.

But read the interpreter's lookup path first:

```c
static c_primitive
lookup_primitive(char *name) {
  /* 1. static table, FIRST */
  for (i = 0; names[i]; i++)
    if (!strcmp(name, names[i]))
      return builtin_cprim[i];
  /* 2. only then: dlsym() */
```

```c
/* generated prims.c */
extern value caml_array_get();
extern value unix_read();
/* ... 1435 lines ... */
c_primitive caml_builtin_cprim[]
  = { caml_array_get,
      unix_read, /*...*/ 0 };
```

Scan every C/C++ source, generate a **superset** `prims.c` (runtime core, unix (131), `str`, bigarray, **655 camlidl** Apron/GMP stubs), disable the `dlopen` branch.

The build montage

library	version	manual trick
OCaml runtime	4.14.2	sak is a *host* tool, rebuild with real cc
GMP	6.1.2	n/a
MPFR	4.2.2	n/a
CamlIDL runtime	latest	3 files, compiles as-is
Apron	lastest	n/a
LLVM/Clang	9	Crosscompilation & adapting the flags
`Clang_to_ml.cc`	n/a	n/a (appart from linking the needed headers)
Camlidl stubs	n/a	Generated C files & compiled them

OCaml 4.14, not 5: the target is wasm32, and OCaml 5 dropped 32-bit.

Several dependencies compiled with **no patch at all**.

The final link

```
emcc ... -o ocamlrun.js \
  --preload-file mopsa.bc@/build/mopsa.bc \
  libs/*.a \
  -s ERROR_ON_UNDEFINED_SYMBOLS=1 \
  prims.o libcamlrun.a
```

- **everything statically linked**: one self-contained 15 MB `.wasm`
- `ERROR_ON_UNDEFINED_SYMBOLS=1`: a missing primitive fails the *link*, loudly, not the browser at runtime
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Every symbol resolved. It links. It's self-contained.

and at runtime, in the browser

```
Uncaught RuntimeError:
index out of bounds
```

Exhibit A: what an OCaml value is

```
     Header                   block data
 ┌──────────────────┐  ┌──────────┬──────────┬────
 │ wosize│ col │ tag│  │ field 0  │ field 1  │ ...
 └──────────────────┘  └──────────┴──────────┴────
       val[-1]            ^ val points here
```

The trap traces to reading a block's *tag* (OCaml 4.14, little-endian):

```
#define Tag_val(val) (((unsigned char *) (val)) [-sizeof(value)])
```

Easy hypotheses, eliminated:

- `sizeof(value)` miscompiled? No: a compile-time constant, always **4**
- a 64/32-bit mismatch? No: the ILP32 config is perfectly consistent

Exhibit A: `-sizeof(value)` is not -4

`sizeof` has type `size_t`, which is **unsigned**:

```
-sizeof(value) = -(size_t)4 = 0xFFFFFFFC        (not -4!)
```

Native 32-bit: `p + 0xFFFFFFFC` is a wrapping 32-bit add $\rightarrow$ exactly `p - 4`.
Upstream OCaml works everywhere, *by wraparound*.

wasm32: two ways to add an offset:

- an explicit `i32.add` wraps modulo 2^{32}: fine
- the **static `offset=N`** immediate of a load is an *unsigned* u32, bounds-checked on the full untruncated value: **no wraparound**

LLVM may fold any *non-negative* constant into `offset=N`. `0xFFFFFFFC` qualifies $\rightarrow$ effective address $\approx$ 4 GiB $\rightarrow$ the bounds check traps.

Exhibit A: the only difference is signedness

```
;; ((unsigned char *) v)
;;      [-sizeof(value)]
(func $tag_old (param i32)
               (result i32)
  local.get 0
  i32.load8_u
    offset=4294967292)
```

folded into the load → traps

```
;; ((unsigned char *) v)
;;      [-(int)sizeof(value)]
(func $tag_fixed (param i32)
                 (result i32)
  local.get 0
  i32.const -4
  i32.add
  i32.load8_u)
```

wrapping add → `p - 4`, fine

The fix: cast to *signed* before negating, since a negative displacement can't be folded into `offset=N`. (re-verified with the project's `emcc` 4.0.22)

The fold is backend luck: `wasi-sdk` clang 18 declines it. The fix removes the bet entirely.

Exhibit B: `va_list` changes shape

Porting to wasm32 also retargeted the *embedded* Clang to 32-bit: the sources MOPSA parses are now typed for a 32-bit ABI.

- **x86-64**: `va_list` is an array → decays to a pointer, handled
- **32-bit**: a scalar `void *`, passed *by reference* to `__builtin_va_start` → an `LValueReferenceType`, a case the type translator never needed before

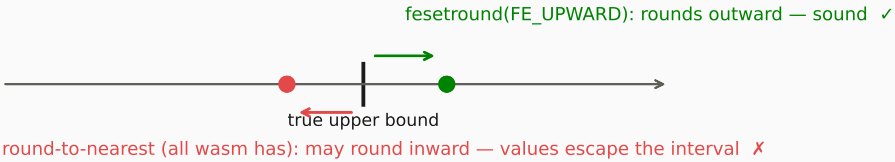
```
unhandled type: lvalue_ref(__builtin_va_list=void*)
```

This broke parsing of the CPython stabs, and with them **all cross C/Python analysis**. The fix (a reference is ABI-equivalent to a pointer):

```
| C.LValueReferenceType tq -> T_pointer (type_qual range tq), no_qual
```

Exhibit C: nobody controls the rounding

Wasm has **no FPU rounding-mode control**: everything rounds to nearest, `fesetround` is a no-op. Not a 32-bit quirk: a property of wasm itself.



- **Apron**: compile every domain with `NUM_MPQ`, so bounds are computed in exact GMP rationals and `fesetround` is never needed. Stub the FPU probe (`-Wl,--wrap=ap_fpu_init`). *Residual*: floats reaching Apron's API.
- **MOPSA's own `floats_round.c`**: *no satisfying fix yet*; widening keeps the analysis sound but coarse.

A fresh instance per analysis

The OCaml runtime keeps its state **global** and is not re-entrant, and `mopsa.bc` ends with `exit`. There is no clean way back.

- keep one instance, suspend around I/O with **Asyncify**? incompatible: OCaml exceptions are `setjmp`/`longjmp`, and both rewrite the stack
- **a fresh instance for every analysis** (`MODULARIZE=1`)

The web worker fetches + `compileStreaming`s the module **once**, then each analysis only *re-instantiates* the already-compiled module.

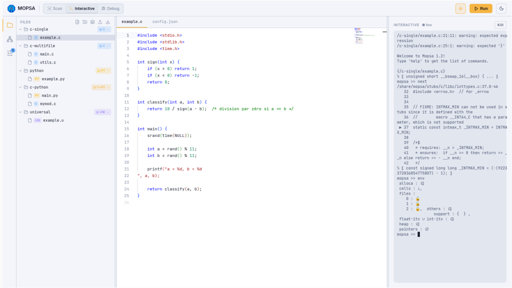
Full module setup (compile, instantiate, unpack the FS data) is **63 ms** under Node, re-instantiation alone is cheaper still.

(Browser: 641 ms for the one-shot first start, fetch included.)

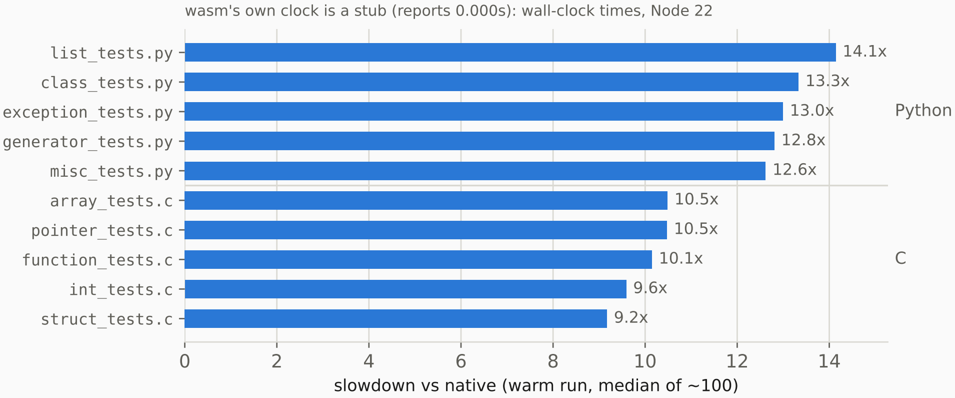
The interactive debugger is a frozen Worker

Interactive REPL & DAP debugger: long-lived runs that **block reading stdin**, inside a Worker that can't service `postMessage` while frozen.

The only web primitive that fits: **`SharedArrayBuffer` + `Atomics.wait`** (needs cross-origin isolation) Cross Origin Embedder Policy and Cross Origin Opener Policy HTTP headers



The cost of interpretation

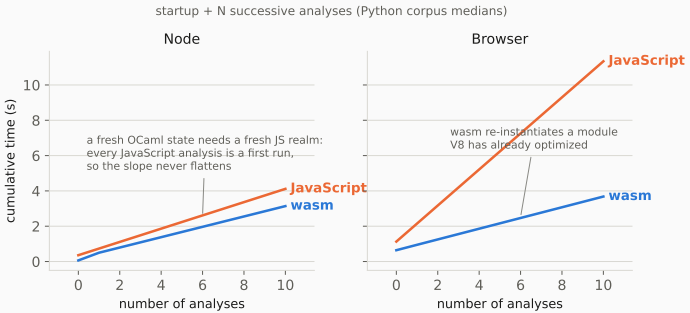


Interpreted bytecode on wasm vs native compiled code, V8 steady state. `int_tests.c` 40 ms → 384 ms

`struct_tests.c` 411 ms → 3.77 s

Comparaison with javascript

the baseline is **Try-MOPSA**, a scaled-down jsoo build: no C frontend, Apron replaced by VPL (pure OCaml).



startup 63 vs 359 ms (Node), 641 ms vs 1.13 s (browser)

repeated Python: wasm ~1.3× (Node) / ~3.4× (browser) ahead

optimistic for jsoo: its 22 MB bundle parse is charged only once

Conclusion

wasm32 is almost a 32-bit target

looked like	actually was	failure mode
bad index / codegen bug	unsigned negation folded into `offset=N`	loud trap
a Clang parser bug	32-bit ABI: `va_list` passed by reference	loud error
nothing at all	no rounding-mode control in wasm	silent unsoundness

The community question

None of this is specific to MOPSA, and most of it is mechanical:

step	
compile project to bytecode	mechanical
compile runtime + deps with emscripten	mostly mechanical (several deps unpatched)
generate `prims.c`, link statically	systematic
genuinely native pieces (LLVM, FFI hazards)	needs care

Worth a shared effort? A dune/opam build target: whole project + native stack → wasm, from an opam switch, at least until a WasmGC backend can interoperate with Emscripten-compiled C. (OCaml 5? `Tag_val` is fixed in 5.x, but 5 dropped 32-bit.)

Try it & thanks

Demo (works right now):

https://mopsawasm.rboud.com/

Code:

https://github.com/rboudrouss/mopsa-emcc

Antoine Miné: guidance through MOPSA and support throughout

Raphaël Monat: Try-MOPSA, the jsoo/VPL baseline

Vincent Chan: `ocaml-wasm` configure tweaks & Unix stubs

Ben Smith (binji): LLVM-to-wasm fork and notes

I'm on the job market, looking for a **CIFRE PhD** host company (French industrial PhD, on static analysis) or an engineering role.

Full benchmark synthesis (~100 reps, medians)

target	files	inst.	cold run	hot run	Mopsa self-time	RSS MB
native	14	n/a	23 ms	n/a	23 ms	70
wasm-node	14	63 ms	476 ms	294 ms	n/a*	193
jsoo-node	9	359 ms	371 ms	n/a**	364 ms	200
wasm-browser	14	641 ms	356 ms	304 ms	n/a*	35***
jsoo-browser	9	1.13 s	1.01 s	1.00 s	298 ms	35***

* the wasm runtime clock is a stub (always 0.000 s) → wall-clock used · ** jsoo-node cannot re-enter a fresh OCaml state in-process · *** browser RSS ≈ main-thread heap only, worker not counted · inst. = compile + instantiate + unpack `.data` (one-shot first start in the browser) · browser “cold” is quasi-hot: the page is shared across files. AMD Ryzen 7 1700X,

Backup: the sak catch

OCaml 4.14 introduced `runtime/sak`, a *host* tool that encodes the stdlib path as a C string. Under `emconfigure` it silently compiles to a `.wasm` that can't run on the host: the path stays empty, failure surfaces at runtime.

```
CFLAGS="$(CFLAGS)" $(EMCONFIGURE) ./configure \
    --disable-native-compiler --disable-systhreads ...
rm -f runtime/sak runtime/sak.o runtime/sak.wasm
cc -c -o runtime/sak.o runtime/sak.c && cc -o runtime/sak runtime/sak.o
touch runtime/sak.o runtime/sak
CFLAGS="$(CFLAGS)" $(MAKE) -C runtime libcamlrun.a
```

`runtime/` also provides the `<caml/*.h>` headers, so every stub is compiled against the exact runtime that executes it.

Backup: LLVM/Clang 9, two-stage build

Stage 1 (native): only `llvm-tblgen` & `clang-tblgen`, with gcc-11 (LLVM 9 no longer compiles with recent gcc/clang).

Stage 2 (wasm): cmake with the Emscripten toolchain, `-DLLVM_TABLEGEN`/`-DCLANG_TABLEGEN` pointing at stage 1.

```
ninja clangFrontend clangParse clangAST clangLex clangBasic
      clangSema clangDriver ... LLVMSupport LLVMCore ...
```

Only the parsing frontend: no codegen backends, no optimizers. Clang's resource headers (`stddef.h`, …) installed then preloaded into the virtual FS; `-DCLANGRESOURCE="/clang-headers"`.

Backup: GMP, MPFR, CamlIDL

- **GMP 6.1.2**: `emconfigure ./configure --disable-assembly --host=none`
- **MPFR 4.2.2**: `touch aclocal.m4 configure` (skip autoconf re-run), `--with-gmp=$(INSTALL_DIR)`
- versions pinned to the pair known to compile cleanly with Emscripten (via a Stack Overflow answer; software archaeology counts)
- **CamlIDL**: runs at *build time* on the host; only its 3-file runtime (`idlalloc.c`, `comintf.c`, `comerror.c`) is compiled to wasm
- **Apron**: each domain (box, oct, polka) as its own archive, `-DNUM_MPQ`

Backup: the va_list patch in full

```
(* References are ABI-equivalent to pointers; model them as such.
   These surface via builtins such as __builtin_va_start, whose
   va_list argument is passed by reference when va_list is a scalar
   (a void pointer) on 32-bit targets, unlike x86-64 where it is an
   array that decays to a pointer. *)
| C.LValueReferenceType tq -> T_pointer (type_qual range tq), no_qual
| C.RValueReferenceType tq -> T_pointer (type_qual range tq), no_qual
```

Trigger: `share/mopsa/stubs/cpython/Python.c`, `PyErr_Format`'s `va_start`; it blocked all cross C/Python analysis on 32-bit.

Backup: versions & availability

- OCaml **4.14.2** (bytecode runtime) · LLVM/Clang **9** · GMP **6.1.2** · MPFR **4.2.2** · emcc **4.0.22**
- artifacts: `ocamlrun.wasm` 15 MB · `ocamlrun.data` 21 MB (virtual FS: `mopsa.bc`, Clang resource headers, linux32 headers, `share/mopsa`)
- fully client-side: https://mopsawasm.rboud.com/
- sources: https://github.com/rboudrouss/mopsa-emcc
- OCaml 5: `Tag_val` fixed upstream in latest 5.x; early experiment suggests the remaining wasm-hostile behavior can be disabled, but OCaml 5 dropped 32-bit support, so wasm32 needs more than that